The Journal of Open Source Software

JURASSIC: A fast infrared radiative transfer model for atmospheric remote sensing

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DOI: 10.xxxxxx/draft

Software

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Editor: Gabriele Bozzola

Reviewers:

- @dhjx1996
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Submitted: 16 April 2026
Published: unpublished

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Summary

The Jülich Rapid Spectral Simulation Code (JURASSIC) is an open source software package for simulating atmospheric radiances and transmittances in the mid-infrared spectral region (approximately 3–20 µm in wavelength). It is designed for atmospheric remote sensing applications that require both physical realism and high computational throughput, such as satellite data analysis, retrieval development and processing, and extensive sensitivity and ensemble studies. JURASSIC supports forward simulations for limb, nadir, zenith, and occultation viewing geometries and includes an integrated optimal estimation framework to retrieve atmospheric state variables.

JURASSIC addresses a common need in atmospheric remote sensing. Line-by-line radiative transfer models provide high spectroscopic accuracy but are often too expensive for large satellite datasets, iterative retrieval workflows, or comprehensive sensitivity studies. JURASSIC instead combines efficient radiative transfer approximations with precomputed emissivity lookup tables, enabling fast calculations while preserving the accuracy required for many research applications. The software is implemented in C, supports OpenMP acceleration together with optional MPI task distribution for retrieval workflows, and is intended for efficient use on multicore workstations and high-performance computing systems.

Statement of need

Infrared remote sensing observations are widely used to derive atmospheric temperature, trace gas abundances, aerosols, and cloud properties from satellite, airborne, balloon-borne, and ground-based instruments. These applications require radiative transfer software that is accurate enough to represent the relevant spectroscopy and viewing geometry, while also being fast enough for large observational datasets and iterative inverse methods.

JURASSIC is designed for researchers working in atmospheric infrared remote sensing, especially those processing large ensembles of atmospheric scenarios, performing sensitivity and uncertainty analyses, or developing retrieval schemes. It provides a general research framework for fast mid-infrared radiative transfer and retrieval that is not restricted to a single instrument or observing configuration.

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State of the field

Atmospheric infrared radiative transfer is supported by a mature ecosystem of software, including line-by-line reference models (Buehler et al., 2018; Clough et al., 2005; Dudhia, 2017) and fast or parameterized forward models (Hocking et al., 2021; Moncet et al., 2008). Comparative studies have examined both classes for satellite sounding and retrieval applications, including radiances, transmittances, and Jacobians (Saunders et al., 2007; Schreier et al., 2018; von Clarmann et al., 2002). These models emphasize different trade-offs between spectroscopic generality, computational cost, supported observing configurations, and ease of adaptation to new applications. JURASSIC occupies a complementary role within this landscape as a research framework for fast thermal-infrared radiative transfer and retrieval across multiple observation geometries.

In addition to forward modelling, JURASSIC includes finite-difference Jacobians and an optimal estimation retrieval framework, allowing radiative transfer and inversion to be performed within the same software environment. JURASSIC is particularly useful when high computational throughput is required for a well-defined instrument or spectral configuration and when flexibility with respect to viewing geometry, atmospheric state, and retrieval setup is important. Its computational efficiency relies on precomputed emissivity lookup tables and radiative transfer approximations. For a new instrument or channel configuration, suitable lookup tables therefore need to be generated and the approximation accuracy should be validated against a line-by-line reference model. Established fast model systems may be preferable for already supported operational sensor configurations, whereas line-by-line models remain the natural choice when maximum spectroscopic generality or reference model accuracy is required.

The core version of JURASSIC described here has been used for limb and nadir radiative transfer applications, including retrievals of atmospheric temperature and trace gas abundances (Hoffmann et al., 2008; Hoffmann & Alexander, 2009). Related extensions, not covered in this paper, have also supported tomographic retrievals for novel limb imaging concepts (Ungermann, Kaufmann, et al., 2010; Ungermann, Hoffmann, et al., 2010), cases affected by clouds and aerosols (Griessbach et al., 2013, 2014, 2016), and applications using GPUs (Baumeister & Hoffmann, 2022). Together, these developments illustrate the use of JURASSIC as a reusable research framework across different infrared remote sensing configurations.

Software design

JURASSIC is organized into modular components for ray tracing, spectroscopy, radiative transfer, and retrieval. This modular design supports code reuse across applications, simplifies maintenance, and makes it easier to adapt individual components to specific research needs. The software is written in C and supports OpenMP acceleration together with optional MPI task distribution for retrieval workflows, enabling efficient use of multicore systems and distributed execution of independent retrieval cases. JURASSIC supports NetCDF input and output for atmospheric data, observation data, and lookup tables, improving interoperability with common scientific data workflows.

The radiative transfer formulation in the core version of JURASSIC described here assumes a vertically stratified atmosphere and curved ray paths that account for atmospheric refraction. The software supports limb, nadir, zenith, and occultation viewing geometries for sensors located inside or outside the atmosphere, corresponding to tangent-path, downward-looking, upward-looking, and transmission-style observation configurations, respectively. Figure 1 shows example mid-infrared limb and nadir radiance spectra simulated with JURASSIC for a midlatitude reference atmosphere, including the contributions of selected trace gases.

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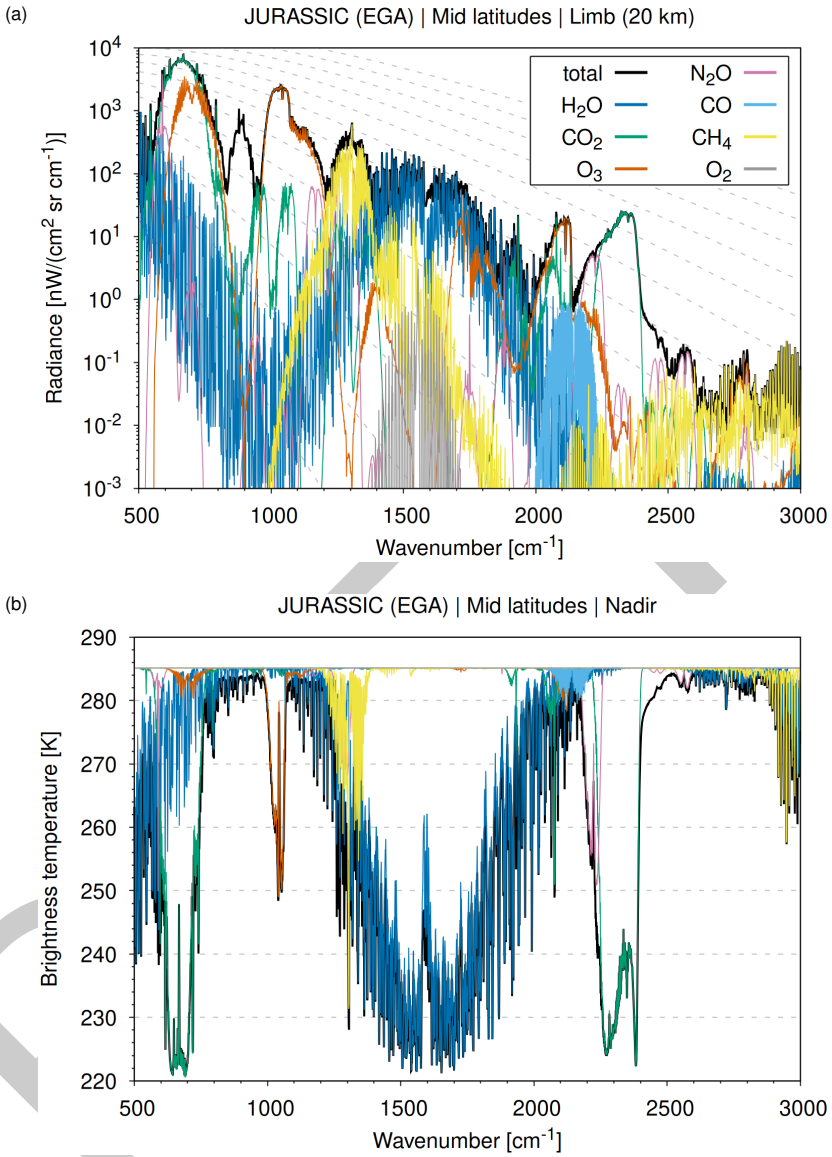


Figure 1: Simulated mid-infrared limb (a) and nadir (b) spectra from JURASSIC for a midlatitude atmosphere at 1 cm^{-1} resolution. Coloured curves show contributions from individual gases, and the black curve shows the total spectrum.

The core model assumes local thermodynamic equilibrium and focuses primarily on thermal infrared radiative transfer in clear air, while also supporting simplified grey-body and extinction-based treatments of clouds and aerosols. At the lower atmospheric boundary, surface temperature and, when configured, spectrally varying emissivity determine an isotropic thermal surface source; optional surface treatments add reflected downwelling thermal radiation and a simplified solar contribution reflected by the surface. The core model does not include solar scattering by cloud and aerosol particles.

To approximate infrared absorption and emission efficiently, JURASSIC uses the Emissivity Growth Approximation (EGA) and the Curtis–Godson Approximation (CGA) (Godson, 1953; Gordley & Russell, 1981; Marshall et al., 1994). JURASSIC applies a band transmittance approach based on spectrally averaged quantities for each instrument channel rather than monochromatic calculations. The precomputed emissivity lookup tables contain emissivities averaged over the spectral response of each channel and are derived from detailed line-by-line

calculations. During runtime, the tables are evaluated by interpolation, enabling inexpensive repeated forward model evaluations. The main trade-offs are the memory required for the lookup tables and the restriction of the calculations to the atmospheric and spectral ranges represented by those tables.

The accuracy and computational performance of JURASSIC have been evaluated in several previous studies. Comparisons with line-by-line calculations for different satellite applications demonstrated radiance accuracy below one percent for typical configurations together with reductions in computational cost of about three orders of magnitude (Hoffmann et al., 2008; Hoffmann & Alexander, 2009). A later comparison with the KOPRA line-by-line model assessed the accuracy of the spectral averaging approach for clear-air and scattering calculations (Griessbach et al., 2013), while more recent work investigated numerical consistency and performance on GPU architectures (Baumeister & Hoffmann, 2022).

For the current implementation, reproducible comparisons with RFM over the 500–3000 cm^{-1} spectral range show median absolute relative limb radiance differences of about 0.1–1.2% and nadir/zenith RMS brightness temperature differences of about 0.4–0.6 K. Depending on viewing geometry and approximation, JURASSIC is roughly two to three orders of magnitude faster than RFM on the recorded reference system using a single thread. OpenMP benchmarks for representative forward model batches on one 64-core AMD EPYC 7742 socket achieve about 50× speedup at 64 threads, corresponding to about 80% parallel efficiency. Detailed benchmark configurations, results, and reproduction instructions are provided in the repository.

The retrieval component implements optimal estimation methods for deriving atmospheric quantities from radiance measurements (Rodgers, 2000). Its state vector can include atmospheric pressure, temperature, trace gas volume mixing ratios, extinction that varies by spectral window, cloud parameters, surface temperature, and spectrally varying surface emissivity. The forward and retrieval components can also be combined for OSSE studies using synthetic radiances. Integrating retrieval capabilities with the forward model reduces duplication of model interfaces, supports reproducible inversion workflows, and facilitates sensitivity studies of forward model assumptions within retrieval applications.

Jacobians required by the retrieval are calculated using finite differences. This approach keeps the derivative calculation general and closely coupled to the forward model, at the cost of additional forward model evaluations for perturbed state vector elements. These calculations are independent and are parallelized with OpenMP. For retrieval workloads containing many independent cases, JURASSIC additionally supports MPI task distribution. Computational resources can therefore be used either for shared memory acceleration of an individual retrieval or for distributing independent retrieval cases, depending on the workload.

A complementary reproducible end-to-end retrieval validation is also provided in the repository. It exercises the finite-difference Jacobians together with the optimal estimation inversion for a synthetic limb retrieval of temperature and trace gases. The reference case converges in three iterations, substantially reduces profile and radiance residual errors, and provides averaging kernels, posterior uncertainties, and runtime diagnostics. This example therefore validates the complete numerical retrieval workflow in addition to the forward model alone.

Related developments outside the scope of this paper include more advanced scattering extensions for cases affected by clouds and aerosols (Griessbach et al., 2014, 2016), an implementation for GPUs on heterogeneous architectures (Baumeister & Hoffmann, 2022), and recent integration efforts aimed at connecting the core model with scattering and accelerator variants in a more maintainable library framework called JURASSIC-UNIFIED (Požgaj, 2022).

Research impact statement

JURASSIC has been used in a broad range of atmospheric remote sensing studies spanning multiple instrument generations and scientific applications. Its early use in limb sounding

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was demonstrated with measurements from the Michelson Interferometer for Passive Atmospheric Sounding (MIPAS), where it supported retrievals and climatological analyses of chlorofluorocarbons and other trace gases (Hoffmann et al., 2008). It was later applied to nadir observations from the Atmospheric Infrared Sounder (AIRS) for temperature retrievals and gravity wave studies (Hoffmann & Alexander, 2009). These applications established JURASSIC as a practical tool for both limb and nadir infrared remote sensing.

Subsequent developments building on JURASSIC expanded its scientific scope. Extensions for aerosols and clouds enabled applications to volcanic aerosol observations and infrared limb measurements affected by clouds (Griessbach et al., 2014, 2016). Tomographic variants supported retrievals of mesoscale atmospheric structures from airborne and satellite observations (Ungermann, Kaufmann, et al., 2010; Ungermann, Hoffmann, et al., 2010; Ungermann et al., 2012). More recent work demonstrated GPU acceleration for efficient execution on heterogeneous architectures (Baumeister & Hoffmann, 2022). Together, these studies show that JURASSIC has served as a reusable framework for radiative transfer and retrieval research across multiple observing geometries and application settings.

The code base includes automated tests and documented example configurations for representative limb, nadir, and zenith applications. Reference cases allow users to verify installation and numerical behaviour, while reproducible validation and performance benchmarks document the accuracy, runtime, and parallel scalability of the current implementation. Public source code hosting, versioned releases, and a persistent Zenodo archive further support citation, reproducibility, and long-term reuse. Together with the scientific applications described above, these resources demonstrate the established use and reproducibility of JURASSIC.

AI usage disclosure

We used ChatGPT (OpenAI, GPT-5.6 Sol) and Codex (OpenAI, GPT-6-Sol) to assist with software development, code review, testing, documentation, and manuscript revision. All AI-assisted outputs were reviewed, edited, and validated by the authors. The authors made all core scientific and software design decisions and remain fully responsible for the submitted software and manuscript.

Acknowledgements

We acknowledge contributions from the atmospheric remote sensing community for testing, validation, and scientific feedback. Computational and storage resources for development and applications of JURASSIC were provided by the Jülich Supercomputing Centre.

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